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Operating Manual

Pressure Transmitter DCT
with RS485 Modbus RTU Interface / i²C Interface

DCT 131, DCT 132, DCT 161, DCT 162,
DCT 531, DCT 531i, DCT 531P, DCT 532, DCT 532i,
DCT 561, DCT 562, DCT 571

DCT 531P:

74-07

READ THOROUGHLY BEFORE USING THE DEVICE
KEEP FOR FUTURE REFERENCE

ID: BA_DCT_i2C-RS485_E | Version: 07.2021.0

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1. General and safety-related information on this operating manual

This operating manual enables safe and proper handling of the product, and forms part of the device. It should be kept in close proximity to the place of use, accessible for staff members at any time.

All persons entrusted with the mounting, installation, putting into service, operation, maintenance, removal from service, and disposal of the device must have read and understood the operating manual and in particular the safety-related information.

Complementary to this operating manual the current data sheet has to be adhered to.

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In addition, the applicable accident prevention regulations, safety requirements, and country-specific installation standards as well as the accepted engineering standards must be observed.

1.1 Symbols Used

Warning word	- Type and source of danger - Measures to avoid the danger

Warning word	Meaning
DANGER	- Imminent danger! - Non-compliance **will result in** death or serious injury.
WARNING	- Possible danger! - Non-compliance **may result in** death or serious injury.
CAUTION	- Hazardous situation! - Non-compliance **may result in** minor or moderate injury.

NOTE - draws attention to a possibly hazardous situation that may result in property damage in case of non-compliance.

✓ Precondition of an action

1.2 Staff qualification

Qualified persons are persons that are familiar with the mounting, installation, putting into service, operation, maintenance, removal from service, and disposal of the product and have the appropriate qualification for their activity.

This includes persons that meet at least one of the following three requirements:

- They know the safety concepts of metrology and automation technology and are familiar therewith as project staff.
- They are operating staff of the measuring and automation systems and have been instructed in the handling of the systems. They are familiar with the operation of the devices and technologies described in this documentation.
- They are commissioning specialists or are employed in the service department and have completed training that qualifies them for the repair of the system. In addition, they are authorized to put into operation, to ground, and to mark circuits and devices according to the safety engineering standards.

All work with this product must be carried out by qualified persons!

1.3 Intended use

The device is intended for converting the physical parameter of pressure into an electric signal. It has to be used only for this purpose, considering the following information.

Devices with 3-A and / or EHEDG certified process connection have been developed especially for applications in food and pharmaceutical industry. The process connection is hygienic and can be sterilized.

Permissible measuring and cleaning media are gases or liquids, which are compatible with the media wetted parts of the device (according to data sheet) and your system. This must be ensured for the application.

The user must check whether the device is suited for the selected use. In case of doubt, please contact our sales department: info@bdsensors.de | phone: +49 (0) 92 35 98 11 0

BD|SENSORS assumes no liability for any wrong selection and the consequences thereof!

The technical data listed in the current data sheet are engaging and must absolutely be complied with. If the data sheet is not available, please order or download it from our homepage: http://www.bdsensors.de

1.4 Incorrect use

WARNING	**Danger through incorrect use** - Only use the device in permissible media and in accordance with its intended use. - Do not use the device as a ladder or climbing aid. - The device must not be altered or modified in any way. - BD|SENSORS is not liable for damage caused by improper or incorrect use.

1.5 Limitation of liability and warranty

Failure to observe the instructions or technical regulations, improper use and use not as intended, and alteration of or damage to the device will result in the forfeiture of warranty and liability claims.

1.6 Safe handling

NOTE - Do not use any force when installing the device to prevent damage of the device and the plant!

NOTE - Treat the device with care both in the packed and unpacked condition!

NOTE - Do not throw or drop the device!

NOTE - Excessive dust accumulation and complete coverage with dust must be prevented!

NOTE - The device is state-of-the-art and is operationally reliable. Residual hazards may originate from the device if it is used or operated improperly.

1.7 Scope of delivery

Check that all parts listed in the scope of delivery are included free of damage, and have been delivered according to your purchase order:

- pressure transmitters
- for DIN 3852 mech. connectors: O-ring (pre-fitted)
- this operating manual

1.8 UL approval (for devices with UL marking)

The UL approval was effected by applying the US standards, which also conform to the applicable Canadian standards on safety.

Observe the following points so that the device meets the requirements of the UL approval:

- only indoor usage
- maximum operating voltage: according to data sheet
- The device must be operated via a supply with energy limitation (acc. to UL 61010) or an NEC Class 2 energy supply.

2. Product identification

The device can be identified by means of the manufacturing label with ordering code. The most important data can be gathered therefrom.

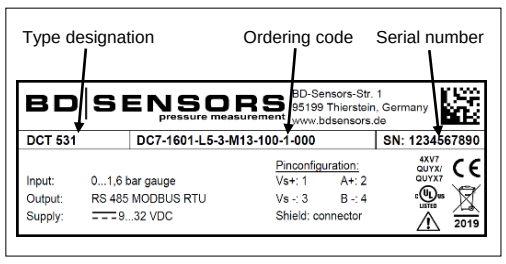


Fig. 1: Example of manufacturing label

NOTE - The manufacturing label must not be removed!

3. Mounting

3.1 Mounting and safety instructions

DANGER	**Danger of death from airborne parts, leaking fluid, electric shock** - Improper installation may result in electric shock! - Always mount the device in a depressurized and de-energized condition!

NOTE - Do not remove the packaging or protective caps of the device until shortly before the mounting procedure, in order to exclude any damage to the diaphragm and the threads! Protective caps must be kept! Dispose of the packaging properly!

NOTE - If there is increased risk of damage to the device by lightning strike or overvoltage, increased lightning protection must additionally be provided!

NOTE - Treat any unprotected diaphragm with utmost care; this can be damaged very easily.

NOTE - Provide a cooling line when using the device in steam piping and and clarify the material compatibility.

NOTE - The measuring point must be designed in such a way that cavitation and pressure surges are avoided.

NOTE - When installing the device, avoid high mechanical stresses on the pressure port! This will result in a shift of the characteristic curve or to damage, in particular in case of very small pressure ranges and devices with a pressure port made of plastic.

NOTE - In hydraulic systems, position the device in such a way that the pressure port points upward (ventilation).

NOTE - If the device is installed with the pressure port pointing upwards, ensure that no liquid drains off on the device. This could result in humidity and dirt blocking the gauge reference in the housing and could lead to malfunctions. If necessary, dust and dirt must be removed from the edge of the screwed joint of the electrical connection.

NOTE - The permissible tightening torque depends on the conditions on site (material and geometry of the mounting point). The specified tightening torques for the pressure transmitter must not be exceeded!

NOTES - for mounting outdoors or in a moist environment:

- Please note that your application does not show a dew point, which causes condensation and can damage the pressure transmitter. There are specially protected pressure transmitters for these operating conditions. Please contact us in such case.
- Connect the device electrically straightaway after mounting or prevent moisture penetration, e.g. by a suitable protective cap. (The ingress protection specified in the data sheet applies to the connected device.)
- Select the mounting position such that splashed and condensed water can drain off. Stationary liquid on sealing surfaces must be excluded!
- If the device has a cable outlet, the outgoing cable must be routed downwards. If the cable needs to be routed upwards, this must be done in an initially downward curve.
- Mount the device such that it is protected from direct solar radiation. In the most unfavourable case, direct solar radiation leads to the exceeding of the permissible operating temperature.
- For devices with gauge reference in the housing (small hole next to the electrical connection), install the device in such a way, that the gauge reference is protected from dirt and moisture. Should the device be exposed to fluid admission, the functionality will be blocked by the gauge reference. An exact measurement in this condition is not possible. Furthermore, this can lead to damages on the device.

3.2 Conditions for devices with 3-A symbol

The device or its connecting piece must be installed in such a way that the surfaces are self-draining (permissible installation position 273° … 87°).

Make sure that the welding socket is mounted flush inside the tank.

The user is responsible for:

- the correct size of the seal and the choice of an elastomeric sealing material that complies with the 3-A standard
- an easy to clean installation position of the pressure transmitter with little dead space, as well as definition / verification / validation of a suitable cleaning process
- defining adequate service intervals

3.3 Conditions for devices, with EHEDG certificate

Install the device according to the requirements given in EHEDG Guidelines 8, 10 and 37. That is to mount the device in a self-draining orientation. The device should be installed flush to the process area. If mounting in a T-piece, the ratio between the depth of the upstand (L) and the diameter (D) of the upstand shall be L/D<1. If welded adapters are used, the food contact surface must be smooth, and the welding has to be done according to EHEDG Guideline 9 and 35. Suitable pipe couplings and process connections must be applied according to the EHEDG Position Paper. (List the available ones.)

3.4 Conditions for oxygen applications

DANGER	**Danger of death from explosion** - when used improperly

Make sure that your device was ordered for oxygen applications and delivered accordingly. (see manufacturing label - ordering code ends with the numbers "007")

Unpack the device directly prior to the installation.

Skin contact during unpacking and installation must be avoided to prevent fatty residues remaining on the device.
Wear safety gloves!

The entire system must meet the requirements of BAM (DIN 19247)!

For oxygen applications > 25 bar, devices without seals are recommended.

Transmitters with o-rings of FKM (Vi 567):
permissible maximum values: 25 bar / 150° C (BAM approval)

3.5 Mounting steps for connections according to DIN 3852

NOTE - Do not use any additional sealing material such as yarn, hemp, or Teflon tape!

✓ The O-ring is undamaged and seated in the designated groove.

✓ The sealing face of the mating component has a flawless surface. (R_Z 3.2)

1 Screw the device into the corresponding thread by hand.

2 Devices equipped with a knurled ring:
only tighten by hand

3 Devices with a spanner flat must be tightened using a suitable open-end wrench. Permissible tightening torques for pressure transmitter:
- Wrench flat made of steel:
  G1/4": approx. 5 Nm G1/2": approx. 10 Nm
  G3/4": approx. 15 Nm G1": approx. 20 Nm
- Wrench flat made of plastic: max. 3 Nm

3.6 Mounting steps for connections according to EN 837

✓ A suitable seal for the medium and the pressure to be measured is available. (e.g. a copper seal)

✓ The sealing face of the mating component has a flawless surface. (R_Z 6.3)

1 Screw the device into the corresponding thread by hand.

2 Then tighten it using an open-end wrench. Permissible tightening torques for pressure transmitter:
G1/4": approx. 20 Nm; G1/2": approx. 50 Nm

3.7 Mounting steps for NPT connections

✓ Suitable fluid-compatible sealing material, e.g. PTFE tape, is available.

1 Screw the device into the corresponding thread by hand

2 Then tighten it using an open-end wrench. Permissible tightening torques for pressure transmitter:
1/4" NPT: approx. 30 Nm; 1/2" NPT: approx. 70 Nm

3.8 Mounting steps for G1″ cone connection

1 Screw the device into the mating thread by hand (seal produced metallically)

2 Then tighten it using an open-end wrench. Permissible tightening torques for pressure transmitter:
p_N < 10 bar: 30 Nm
p_N ≥ 10 bar: 60 Nm

3.9 Mounting steps for Clamp and Varivent® connections

✓ A suitable seal for the measured fluid and the pressure to be measured is available.

✓ Chapter "3.2 and/or 3.3" have been noticed.
EHEDG conformity is only ensured in combination with an approved seal. This is e.g.:
for Clamp connections - codes C61, C62, C63:
T-ring seal from Combifit International B.V.
for Varivent® connections - codes P40, P41:
EPDM-O-ring which is FDA-listed
Note, that P40 can only be used for tank flanges.

1 Place the seal onto the corresponding mounting part.

2 Centre the clamp connection or Varivent® connection above the counterpart with seal.

3 Then fit the device with a suitable fastening element (e. g. semi-ring or retractable ring clamp) according to the supplier's instructions.

4. Electrical connection

4.1 Connection and safety instructions

DANGER	**Danger of death from electric shock** - Always mount the device in a depressurized and de-energized condition!

✓ The supply corresponds to protection class III (protective insulation).

NOTE - Use a shielded and twisted multicore cable for the electrical connection.

NOTE - for devices with **cable outlet**

- When routing the cable, following bending radiuses have to be complied with:
  cable without ventilation tube:
  static installation: 8-fold cable diameter
  dynamic application: 12-fold cable diameter
  cable with ventilation tube:
  static installation: 10-fold cable diameter
  dynamic application: 20-fold cable diameter
- In case of devices with **cable outlet** and integrated ventilation tube, the PTFE filter located at the cable end on the air tube must neither be damaged nor removed! Route the end of the cable into an area or suitable connection box which is as dry as possible and free from aggressive gases, in order to prevent any damage.

4.2 Electrical installation

Establish the electrical connection of the device according to the technical data shown on the manufacturing label, the following table and the wiring diagram.

Pin configuration:

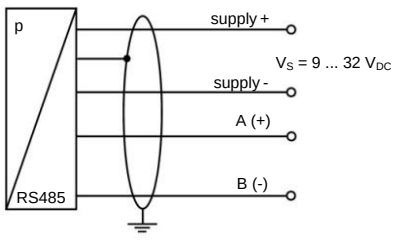
Interface	Electrical connections	M12x1 (4-pin) metal	Binder 723 (5-pin)	cable colours (IEC 60757)
RS 485 Modbus RTU	Supply +	1	1	WH (white)
	Supply –	3	3	BN (brown)
	not inverted A +	2	2	GN (green)
	inverted B –	4	4	YE (yellow)
	Shield	plug housing	plug housing	GNYE (green-yellow)

Interface	Electrical connections	M12x1, metal (5-pin)
RS 485 Modbus RTU with reset function for DCT 531 DCT 531i DCT 531P DCT 561 DCT 571	Supply +	1
	Supply –	3
	not inverted A +	2
	inverted B –	4
	Reset	5
	Shield	plug housing

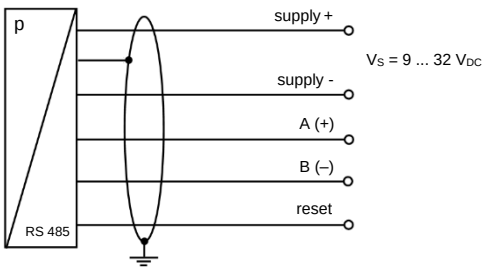
Interface	Electrical connections	M12x1 (5-pin) metal	Binder 723 (5-pin)	cable colours (IEC 60757)
I²C	Supply+	1	1	WH (white)
	Supply –	3	3	BN (brown)
	SDA	2	2	YE (yellow)
	SCL	4	4	GN (green)
	INT	5	5	PK (pink)
	Shield	plug housing	plug housing	GNYE (green-yellow)

Wiring diagrams:

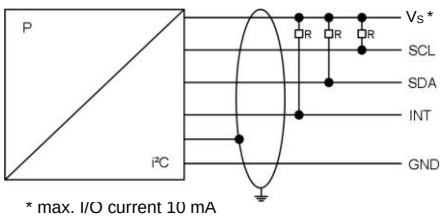
RS 485 / Modbus RTU



RS 485 / Modbus RTU with reset function
for DCT 531, DCT 531i, DCT 531P, DCT 561 und DCT 571



I²C



* max. I/O current 10 mA

5. Commissioning

DANGER	**Danger of death from airborne parts, leaking fluid, electric shock** - Operate the device only within the specification! (according to data sheet)

✓ The device has been installed properly.

✓ The device does not have any visible defect.

6. Modbus RTU communication

6.1 Configuration of Modbus RTU

✓ Delay time (start-up time) of 500 msec has been considered.

[ ] concerns only DCT 531i

Factory setting	1	1	1
Address	1 … 247		
Baud-Rate			
4800		0	
9600		1	
19200		2	
38400		3	
Paritate			
none			0
odd			1
even			2

6.2 Explicit register description

Map of Input registers (read only, function #4 - Read Input Registers)		
Address	**Register**	**Data type**
0x0000	Serial number	UInt32
0x0001		
0x0002	Date of last calibration	Date
0x0003		
0x0004	Upper range of pressure channel	Float, IEEE754
0x0005		
0x0006	Lower range of pressure channel	Float, IEEE754
0x0007		
0x0008	Actual pressure	Float, IEEE754
0x0009		
0x000A	Maximal pressure	Float, IEEE754
0x000B		
0x000C	Minimal pressure	Float, IEEE754
0x000D		

Map of Input registers (read only, function #4 - Read Input Registers)		
Address	**Register**	**Data type**
0x000E	Upper range of temperature channel	Float, IEEE754
0x000F		
0x0010	Lower range of temperature channel	Float, IEEE754
0x0011		
0x0012	Actual temperature	Float, IEEE754
0x0013		
0x0014	Maximal temperature	Float, IEEE754
0x0015		
0x0016	Minimal temperature	Float, IEEE754
0x0017		

Map of Holding registers (read, write, fce #3 - Read Holding Registers , fce #6 - Write Single Register)		
Address	**Register (description)**	**Data type**
0x0000	Unit of pressure channel	Uint16
0x0001	Unit of temperature channel	Uint16
0x0002	Device address	Uint16
0x0003	Baud rate	Uint16
0x0004	Parity	Uint16

Pressure unit enumeration	
Code (Uint16)	**Unit**
0x0003	mmH_2O
0x0004	mmHG
0x0005	psi
0x0006	bar
0x0007	mbar
0x0008	g/cm²
0x0009	kg/cm²
0x000A	Pa
0x000B	kPa
0x000C	torr
0x000D	atm
0x000E	mH_2O
0x000F	MPa

Temperature unit enumeration	
Code (Uint16)	**Unit**
0x0000	°C
0x0001	°K
0x0002	°F

Baud rate enumeration	
Code (Uint16)	**Baud rate [Bd]**
0x0004	4800
0x0005	9600
0x0006	19200
0x0007	38400

Parity enumeration	
Code (Uint16)	**Parity**
0x0000	none
0x0001	odd
0x0002	even

6.3 Reset function for DCT 531, DCT 531i, DCT 531P, DCT 561 and DCT 571

The reset function of the device is a service function that is not used in regular operation (bus operation). The reset function is used to reset the device to the factory settings, which is extremely helpful, especially with the bus address. In normal operation, no signal may be connected with the reset input.

Activation of the reset function:

To reset the device to the factory settings, the reset input must be connected to 24 V at the same time as the positive supply voltage input. The bus signals A + B can be connected or not connected at this time.

After connecting 24 V once, the reset was carried out internally and the reset input must be disconnected again.

7. i²C-Interface

7.1 Configuration of i²C-interface

Concerns only DCT 532i

Factory setting	050	0	0	0	0	00001
Slave address						
address	1 … 127					
Type of result register						
32bit IEEE float		0				
16bit integer		1				
Byte order of values						
Low byte first			0			
High byte first			1			
Mode of result register						
Value				0		
Percent of nominal				1		
Restore of address pointer						
no restore					0	
to last set address on next start					1	
Digital meaning						
Count of result						00001 … 10000

7.2 Register overview

Register	Type 0 (Float)	Type 1 (Int 16)
0x00	Status	Status
0x01	Pressure	Pressure
0x02		
0x03		Temperature
0x04		
0x05	Temperature	
0x06		
0x07		
0x08		
0x40	Configuration	Configuration
0x41	Oversampling	Oversampling
0x42		
0x43	Slave Address	Slave Address
0x44	Pressure unit	Pressure unit
0x45	Nominal pressure lower	Nominal pressure lower
0x46		
0x47		Decimal places
0x48		
0x49	Nominal pressure upper	Nominal pressure upper
0x4A		
0x4B		
0x4C		
0x4D	Temperature unit	Temperature unit
0x4E	Nominal temperature lower	Nominal temperature lower
0x4F		
0x50		Decimal places
0x51		
0x52	Nominal temperature upper	Nominal temperature upper
0x53		
0x54		
0x55		

7.3 Explicit register description

Explanation:

r = only readable

r/w = read and write capable

d = don´t care

0x00 – Status register:

7	6	5	4	3	2	1	0
ABS			ERR	SAT	OVER	UNDER	READY
r	d	d	r	r	r	r	r

bit 0	Result registers is **READY**	
	0 b =	Outdated values will be read
	1 b =	Registers contain new values
	Note:	This bit has same behaviour as hardware ready connector. Logic level is inverted because of open collector at output stage.
	Note:	It is possible to poll update without using hard wiring, or to check which sensor has updated if more than one is used on bus.
bit 2	Value is out of **UNDER** nominal range	
	0 b =	Pressure value is in nominal range
	1 b =	Pressure is to low
	Note:	OVER and UNDER flags are stored until state register is read.

bit 3	Value **SAT**urated	
	0 b =	No saturation
	1 b =	Output value or ADC is out of range
bit 4	Internal **ERR**or, transmitter does not work	
	0 b =	Transmitter is in normal operation
	1 b =	Internal error or wrong setting is active
bit 7	Transmitter is **ABS**olute	
	0 b =	Pressure type of transmitter is relative
	1 b =	Pressure type of transmitter is absolute

0x40 – Configuration register

7	6	5	4	3	2	1	0
ADD			RESTORE	MODE		ORDER	TYPE
r/w	d	d	r/w	r/w		r/w	r/w

bit 0	**TYPE** of result register	
	0 b =	32bit IEEE float
	1 b =	16bit integer
bit 1	Byte **ORDER** of values	
	0 b =	Low byte first
	1 b =	High byte first
bit 2…3	**MODE** of result register	
	00b=	Value
	01b=	Percent of nominal
	10b=	reserved
	11b=	reserved
bit 4	**RESTORE** address pointer	
	0 b =	No restore
	1 b =	Restore to last set address on restart
	Note:	Using this setting causes reset of register pointer to last written after each stop condition of readout.
bit 7	Set new I2C slave **ADD**ress	
	0 b =	Slave address stays as it is
	1 b =	Set this bit to apply previously set slave address

0x43 – Slave address register

7	6	5	4	3	2	1	0
SLAVE_ADDRESS							
r/w							d

bit 1…7	**SLAVE ADDRESS** which this transmitter acknowledges	
	Note:	To apply new address, it is necessary to set ADD bit in configuration register after new address is written.

0x44 – Pressure unit register

7	6	5	4	3	2	1	0
UNIT							
r/w							

bit 0…7	Pressure **UNIT** (according to units in HART protocol)	
	0x01	inH2O @ 68°F
	0x02	inHg @ 0°C
	0x03	ftH2O @ 68°F
	0x04	mmH2O @ 68°F
	0x05	mmHG @ 0°C
	0x06	psi
	0x07	bar
	0x08	mbar
	0x09	g/cm²
	0x0A	kg/cm²
	0x0B	Pa
	0x0C	kPa
	0x0D	Torr
	0x0E	atm
	0x91	inH2O @ 60°F
	0xAA	cmH2O @ 4°C
	0xAB	mH2O @ 4°C
	0xAC	cmHg @ 0°C
	0xAD	lb/ft²
	0xAE	hPa
	0xB0	kg/m²
	0xB1	ftH2O @ 4°C
	0xB2	ftH2O @ 60°F
	0xB3	mHg @ 0°C
	0xED	Mpa
	0xEE	inH2O @ 4°C
	0xEF	mmH2O @ 4°C

0x4d – Temperature unit register

7	6	5	4	3	2	1	0
UNIT							
r/w							

bit 0…7	Temperature **UNIT** (according to units in HART protocol)	
	0x20	°C
	0x21	°F
	0x22	°R
	0x23	K
	Note:	If pressure or temperature unit is set to an invalid value, slave will not acknowledge.
	Note:	If 16 bit integer mode is selected and nominal values can not be displayed with 0…5 decimal places, ERROR flag is set and DECIMAL_PLACES will be 0xff.

0x47 / 0x50 – Decimal places register

7	6	5	4	3	2	1	0
DECIMAL_PLACES							
r							

bit 0…7	Count of **DECIMAL_PLACES**	
	Note:	Available only when 16bit integer type is selected.
	Note:	Value will be calculated automatically according to nominal range.

8. Maintenance

DANGER	**Danger of death from airborne parts, leaking fluids, electric shock** - Always service the device in a depressurized and de-energized condition!
WARNING	**Danger of injury from aggressive fluids or pollutants** - Depending on the measured medium, this may constitute a danger to the operator. - Wear suitable protective clothing e.g. gloves, safety goggles.

If necessary, clean the housing of the device using a moist cloth and a non-aggressive cleaning solution.

During the cleaning processes, note the compatibility of the cleaning media used in combination with the media-wetted materials of the pressure measuring devices. Permissible concentrations and temperatures must be observed. Verification/ validation by the user is essential.

For EHEDG certified devices in tanks, the cleaning device must be positioned in such a way that the sensor is directly assessed and wetted for cleaning. The device has been developed for Cleaning in Place (CIP) applications and must not be dismantled for cleaning.

Deposits or contamination may occur on the diaphragm/ pressure port in case of certain media. Depending on kind and quality of the process, suitable cyclical maintenance intervals must be specified by the operator. As part of this, regular checks must be carried out regarding corrosion, damage of diaphragm/seal(s) and signal shift. A periodical replacement of the seal(s) may be necessary.

If the diaphragm is calcified, it is recommended to send the device to BD|SENSORS for decalcification. Please note the chapter "Service / repair" below.

NOTE - Wrong cleaning or improper touch may cause an irreparable damage on the diaphragm. Therefore, never use pointed objects or pressured air for cleaning the diaphragm.

9. Troubleshooting

DANGER	**Danger of death from airborne parts, leaking fluids, electric shock** - If malfunctions cannot be resolved, put the device out of service (proceed according to chapter 10 up to 12)

In case of malfunction, it must be checked whether the device has been correctly installed mechanically and electrically. Use the following table to analyse the cause and resolve the malfunction, if possible.

Fault: no output signal	
Possible cause	**Fault detection / remedy**
Connected incorrectly	Checking of connections
Conductor/wire breakage	Checking of all line connections.
Defective measuring device (signal input)	Checking of ammeter (miniature fuse) or of analogue input of your signal processing unit

Fault: incorrect signal behaviour	
Possible cause	**Fault detection / remedy**
Load resistance too high	Checking of load resistance (value)
Supply voltage too low	Checking of power supply output voltage
Defective energy supply	Checking of the power supply and the supply voltage being applied to the device
Diaphragm of senor is severely contaminated or damaged	Checking of diaphragm; if necessary, send the device to BD|SENSORS for repair

Fault: wrong or no output signal	
Possible cause	**Fault detection / remedy**
Cable damaged mechanically, thermally or chemically	Checking of cable; pitting corrosion on the stainless-steel housing as a result of damage on cable; when damaged, send the device to BD|SENSORS for repair

10. Removal from service

DANGER	**Danger of death from airborne parts, leaking fluids, electric shock** - Disassemble the device in a depressurized and de-energized condition!
WARNING	**Danger of injury from aggressive media or pollutants** - Depending on the measured medium, this may constitute a danger to the operator. - Wear suitable protective clothing e.g. gloves, goggles.

NOTE - After dismounting, mechanical connections must be fitted with protective caps.

11. Service / repair

Information on service / repair:

- www.bdsensors.de
- info@bdsensors.de
- Service phone: +49 (0) 92 35 98 11 0

11.1 Recalibration

During the life-time of a transmitter, the value of offset and span may shift. As a consequence, a deviating signal value in reference to the nominal pressure range starting point or end point may be transmitted. If one of these two phenomena occur after prolonged use, a recalibration is recommended to ensure furthermore high accuracy.

11.2 Return

WARNING	**Danger of injury from aggressive media or pollutants** - Depending on the measured medium, this may constitute a danger to the operator. - Wear suitable protective clothing e.g. gloves, goggles.

Before every return of your device, whether for recalibration, decalcification, modifications or repair, it has to be cleaned carefully and packed shatter-proofed. You have to enclose a notice of return with detailed defect description when sending the device. If your device came in contact with harmful substances, a declaration of decontamination is additionally required.

Appropriate forms can be downloaded from our homepage. Download these by accessing www.bdsensors.de or request them:

info@bdsensors.de | phone: +49 (0) 92 35 / 98 11 0

In case of doubt regarding the fluid used, devices without a declaration of decontamination will only be examined after receipt of an appropriate declaration!

12. Disposal

WARNING	**Danger of injury from aggressive media or pollutants** - Depending on the measured medium, this may constitute a danger to the operator. - Wear suitable protective clothing e.g. gloves, goggles.

The device must be disposed of according to the European Directive 2012/19/EU (waste electrical and electronic equipment). Waste equipment must not be disposed of in household waste!

NOTE - Dispose of the device properly!

13. Warranty terms

The warranty terms are subject to the legal warranty period of 24 months, valid from the date of delivery. If the device is used improperly, modified or damaged, we will rule out any warranty claim. A damaged diaphragm will not be accepted as a warranty case. Likewise, there shall be no entitlement to services or parts provided under warranty if the defects have arisen due to normal wear and tear.

14. EU declaration of conformity / CE

The delivered device fulfils all legal requirements. The applied directives, harmonised standards and documents are listed in the EC declaration of conformity, which is available online at: http://www.bdsensors.de.
Additionally, the operational safety is confirmed by the CE sign on the manufacturing label.